Model Validation and Analysis

Module 4 - ABE 721: Agricultural Systems Modelling and Simulation

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Lecture Outline

1. 1. Epistemology and Metrics of Model Validation
2. 2. Stochastic Models and Uncertainty
3. 3. System Dynamics: Stocks and Flows
4. 4. Causal Loop Diagrams and Archetypes
5. 5. Equilibrium, Stability, and Growth
6. 6. Hydrological Networks (SPAC)

Verification vs. Validation

A calibrated mathematical model is useless if its predictive capabilities cannot be verified against physical reality.

Verification

- *"Are we building the model right?"*
- Ensures the computer code correctly solves the mathematical equations (e.g., RK4 algorithm is bug-free).

Validation

- *"Are we building the right model?"*
- Ensures the equations actually represent the real world.
- **Crucial Rule:** Must use an *independent* dataset not used in calibration.

Quantitative Metrics: RMSE and MAE

Visual inspection is insufficient. We use statistical indices to quantify error.

1. Root Mean Square Error (RMSE)

- Measured in the exact same physical units as the state variable.
- Highly sensitive to large outliers due to the squaring of errors.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(Y_{obs,i} - Y_{sim,i}\right)^2}$$

2. Mean Absolute Error (MAE)

- Linear score where all individual differences are weighted equally.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|Y_{obs,i} - Y_{sim,i}\right|$$

Quantitative Metrics: Nash-Sutcliffe Efficiency (NSE)

Nash-Sutcliffe Efficiency (NSE)

Normalizes the model's accuracy relative to the mean of the observed data ($\bar{Y}_{obs}$). Crucial in hydrological modelling.

$$NSE = 1 - \frac{\sum_{i=1}^{n}(Y_{obs,i} - Y_{sim,i})^2}{\sum_{i=1}^{n}(Y_{obs,i} - \bar{Y}_{obs})^2}$$

Interpreting NSE:

- **NSE = 1.0:** Flawless, perfect match.
- **NSE = 0.0:** The model is only as accurate as simply guessing the historical average.
- **NSE < 0.0:** The model is fundamentally flawed and performs worse than the historical average.

Quantitative Metrics: Bias and Agreement

Percent Bias (PBIAS)

- Measures the tendency of the simulated data to systematically overestimate (-) or underestimate (+) reality.

$$PBIAS = \left[\frac{\sum(Y_{obs,i} - Y_{sim,i})}{\sum Y_{obs,i}} \right] \times 100\%$$

Index of Agreement (d)

- Standardized measure (0 to 1) of the degree of model prediction error. Can detect additive and proportional differences in means and variances.

Why Deterministic Models Fail

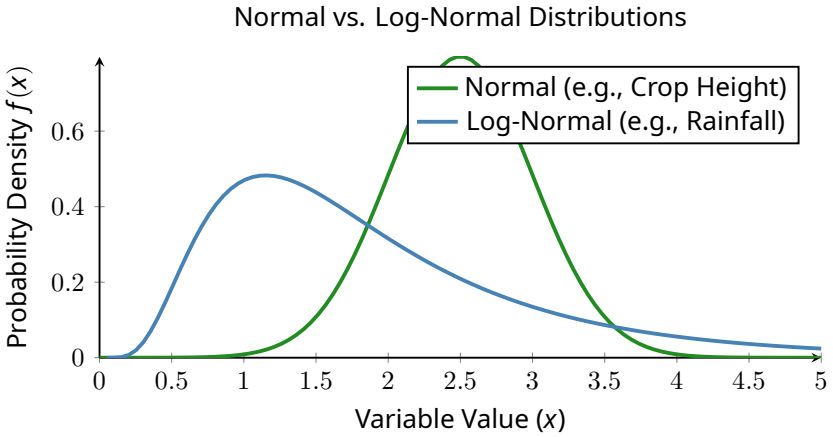
- Agricultural processes are driven by weather, biology, and human intervention—all of which are inherently **chaotic**.
- A deterministic model yields a single, exact numerical output, which fails to capture the true risk reality of farming.
- **Stochastic Models** incorporate probability distributions to generate a *range* of possible outcomes with confidence intervals.

Common Probability Distributions

Inputs (rainfall, wind, machine breakdowns) are treated as random variables mapped to Probability Density Functions (PDFs).

- **Normal (Gaussian):** Symmetric. Used for sensor errors or genetic crop heights.
- **Log-Normal:** Heavily right-skewed and cannot be negative. The global standard for hydrological variables like *daily precipitation*.
- **Weibull:** The standard for modelling stochastic wind speeds.

Comparing Agricultural PDFs



Monte Carlo Simulation Methods

The standard computational technique for stochastic risk analysis.

The Algorithm

1. Define precise probability distributions for all uncertain inputs.
2. Randomly sample one value from every defined distribution.
3. Run the deterministic ODE simulation using these sampled values to generate one output instance (e.g., Yield).
4. Repeat iteratively for thousands of iterations ($N \approx 10,000$).
5. Statistically analyze the aggregate distribution of outputs.

Latin Hypercube Sampling (LHS)

Limitation of standard Monte Carlo: It requires millions of iterations to properly sample extreme ”tails” (e.g., a 1-in-100 year flood).

- **LHS** is an advanced stratified sampling method.
- Divides the cumulative probability distribution into *N* equal intervals.
- Ensures exactly one sample is randomly drawn from *every single interval*.
- Guarantees the entire spectrum of possibilities is tested uniformly, drastically reducing computational time.

Introduction to System Dynamics (SD)

Pioneered by Jay Forrester (MIT), SD provides a rigorous visual and mathematical language for framing complex, non-linear systems.

1. **Stocks (Accumulations)**
   - State variables of the system.
   - If time froze, stocks would still have a measurable value.
   - *Visual:* Solid rectangles.
   - *Ex:* Soil nitrate, water volume.

2. **Flows (Rates)**
   - The active rate of change of stocks.
   - If time freezes, flow is zero.
   - *Visual:* Pipes with valves.
   - *Ex:* Evaporation rate, leaching rate.

Mathematical Integration of SD

The visual representation of SD maps perfectly to integral calculus.

Integral Formulation

$$Stock(t) = Stock(t_0) + \int_{t_0}^{t} \left(\sum Inflows(s) - \sum Outflows(s) \right) ds$$

Taking the derivative with respect to time yields the standard ODE state-space formulation explored in Module 2:

$$\frac{d(Stock)}{dt} = \sum Inflows - \sum Outflows$$

SD Case Study: Soil Nitrogen Cycle

Consider the complex network of the soil nitrogen cycle. We define three primary Stocks:

1. **Organic Nitrogen (N_{org}):** In plant residue and humus.
2. **Ammonium (NH_4^+):** Available to plants, immobile.
3. **Nitrate (NO_3^-):** Available to plants, highly mobile.

Flows governing these stocks:

- Mineralization ($N_{org} \rightarrow NH_4^+$)
- Nitrification ($NH_4^+ \rightarrow NO_3^-$)
- Leaching, Denitrification, and Plant Uptake (Outflows)

Nitrogen Cycle ODE Formulation

The visual map translates directly into a coupled system of ODEs:

$$\frac{dN_{org}}{dt} = Inputs_{residue} - F_{min}$$
$$\frac{dNH_4^+}{dt} = Inputs_{fert} + F_{min} - F_{nit} - F_{up1}$$
$$\frac{dNO_3^-}{dt} = F_{nit} - F_{den} - F_{lea} - F_{up2}$$

By assigning mathematical rate equations to each flow, engineers simulate nutrient availability and optimize fertilizer schedules.

Causal Loop Diagrams (CLDs)

While Stock/Flow diagrams provide quantitative architecture, **CLDs** map the conceptual, qualitative feedback mechanisms driving behavior.

Components:

- **Variables:** Quantities that can vary (Soil Moisture, Yield).
- **Causal Links:** Directed arrows.
- **Polarities (+ or -):**
  - **Positive (+):** Variables move in the *same* direction.
  - **Negative (-):** Variables move in *opposite* directions.
- **Delays (//):** Time delays causing system oscillation.

Feedback Loops in Agriculture

Systems are governed by closed, self-referential loops of causality.

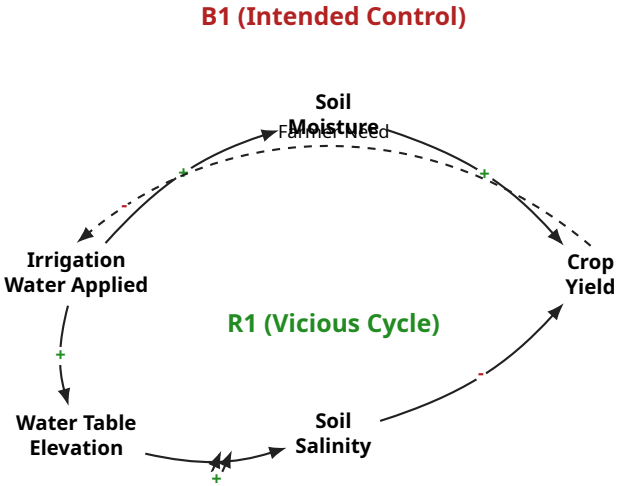
1. Reinforcing Loops (R or +)

Contains an *even* number of negative links. Compounds change, leading to exponential growth or collapse (e.g., Pest outbreaks).

2. Balancing Loops (B or -)

Contains an *odd* number of negative links. Seeks equilibrium and stabilizes the system (e.g., Stomatal closure during drought).

Case Study: Irrigation & Soil Salinity Paradox



Interpreting the Salinity Paradox

- The farmer applies water to increase yield (Balancing Loop B1).
- However, heavy irrigation raises the water table, eventually drawing salts into the root zone via capillary action.
- This increases salinity, decreasing yield, which prompts the farmer to apply *more* water.
- This triggers a vicious **Reinforcing Loop (R1)** of environmental degradation.
- **Engineering Solution:** Break the causal link physically (e.g., install subsurface tile drainage).

Equilibrium Points in Systems

Understanding a system's long-term fate requires finding its **Equilibrium Points** (steady states).

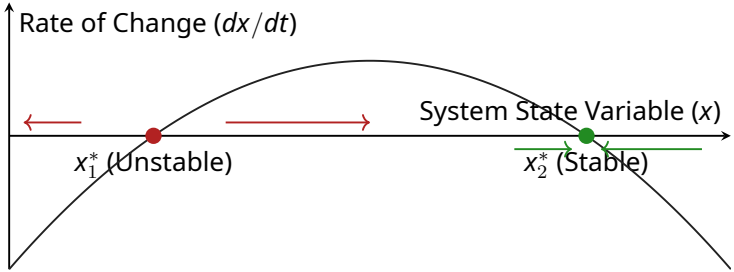
For a dynamic system $\frac{d\mathbf{x}}{dt} = f(\mathbf{x})$, an equilibrium state $\mathbf{x}^*$ exists if and only if:

$$f(\mathbf{x}^*) = 0$$

Stability Classification:

- **Stable (Attractor):** System naturally pulls back to $\mathbf{x}^*$ after a perturbation (Balancing loops dominate).
- **Unstable (Repeller):** System accelerates away from $\mathbf{x}^*$ after a perturbation (Reinforcing loops dominate).

Phase Plane Analysis (1D)



Jacobian Stability Proof (Lotka-Volterra)

To prove stability in multi-variable systems, we use the **Jacobian Matrix** ($\mathbf{J}$) evaluated at $\mathbf{x}^*$.

For the Lotka-Volterra predator-prey model, the Jacobian yields eigenvalues:

$$\lambda = \pm i\sqrt{\alpha\gamma}$$

Stability Theorem

If the real parts of *all* eigenvalues are strictly negative, the system is asymptotically stable.

Here, the real part is zero (purely imaginary). The equilibrium is a *Center*. Pests and predators oscillate infinitely; pests are controlled but never permanently eradicated.

Advanced Models of S-Shaped Growth

Sigmoidal growth describes crop canopy expansion, disease spread, and technology adoption.

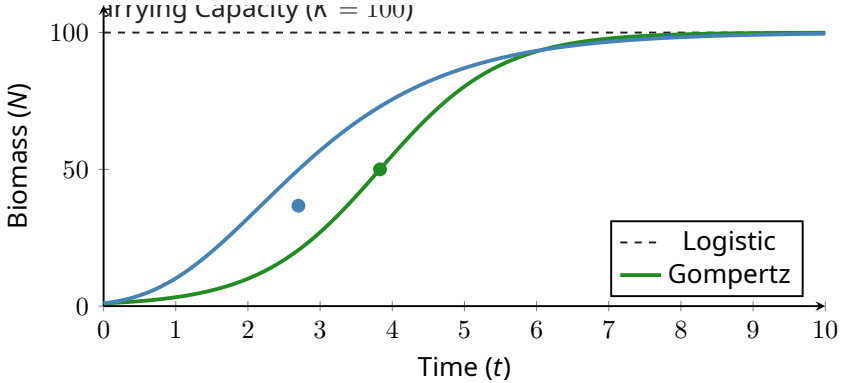
Shift in Loop Dominance:

1. **Exponential Growth:** Reinforcing loop dominates (acceleration).
2. **Inflection Point:** Environmental limits hit. Maximum growth rate.
3. **Asymptotic Goal-Seeking:** Balancing loop dominates, slowing growth to approach Carrying Capacity (K).

Mathematical Formulations of Growth

1. **Logistic Growth (Symmetric)**
   - Inflection point occurs exactly at $N = K/2$.
2. **Gompertz Growth (Asymmetric)**
   - Exponential growth rate decays exponentially.
   - Inflection point occurs much earlier at $N = K/e \approx 0.37K$.
3. **Richards Curve**
   - Includes a shape parameter (ν) for ultimate calibration flexibility.

Comparing Growth Archetypes



Soil-Plant-Atmosphere Continuum (SPAC)

Water movement in farms is modeled as a continuous dynamic flow driven by thermodynamic potential energy gradients.

Potential Gradient Rule

Water flows strictly from regions of high water potential (Ψ) to low potential.

$$\Psi_{soil} > \Psi_{root} > \Psi_{stem} > \Psi_{leaf} > \Psi_{atmosphere}$$

Electrical Resistance Analogy

Flow (q) through SPAC is modelled identically to electrical current (Ohm's Law). Potential difference ($\Delta\Psi$) is voltage; biological/physical impediments are resistance (R).

$$q = \frac{\Delta\Psi}{R_{total}} = \frac{\Psi_{soil} - \Psi_{root}}{R_{soil}} = \frac{\Psi_{root} - \Psi_{leaf}}{R_{plant}} = \frac{\Psi_{leaf} - \Psi_{atm}}{R_{stomata} + R_{boundary}}$$

- If soil dries, R_{soil} increases.
- If Ψ_{leaf} drops dangerously low, plants close stomata ($R_{stomata} \to \infty$), halting the flow to prevent desiccation.

Evapotranspiration: Penman-Monteith

The ultimate outflow driver of SPAC into the atmosphere. The **FAO-56 Penman-Monteith Equation** combines thermodynamic energy balance with aerodynamic mass transfer:

$$ET_0 = \frac{\Delta(R_n - G) + \rho_a c_p \frac{(e_s - e_a)}{r_a}}{\lambda\left(\Delta + \gamma\left(1 + \frac{r_s}{r_a}\right)\right)}$$

- R_n: Net solar radiation.
- $(e_s - e_a)$: Vapor Pressure Deficit (VPD).
- r_a: Aerodynamic resistance (wind).
- r_s: Stomatal surface resistance (biological control).

Summary of Module 4

- **Validation:** We must use quantitative metrics (NSE, RMSE) and independent data to verify model accuracy.
- **Stochasticity:** Monte Carlo and LHS integrate probability distributions to capture weather/biological uncertainty.
- **System Dynamics:** Causal loops identify systemic archetypes (vicious cycles), and Stocks/Flows define the calculus.
- **Stability & Flow:** Jacobian matrices prove equilibrium stability, while potential gradients drive SPAC water flows.

End of Module 4

Questions & Discussion